Dataset GRINS and resilience

GOAL: evaluating the resilience of Italian municipalities, by reprojecting the GDP at municipal level with Fixed-Rank Kriging (FRK).

STEP 1: Introduction

This work aims to evaluate the **resilience** of Italian municipalities by constructing a **Social Growth Index (SGI)** and providing a comparative ranking. The selected variables for measuring resilience are the growth rate of GDP, GDP per capita, and population over the period 2015–2022. Due to the absence of municipal-level GDP data, estimates were produced using Fixed-Rank Kriging (FRK). Spatial dependence between municipalities was assessed using LISA Maps, which highlight clusters and local patterns of similarity across the three dimensions. Municipal resilience was then ranked using the Copeland method, which compares performance through pairwise comparisons. To ensure stability over time, the method includes a spatial adjustment that incorporates information from neighbouring municipalities. This approach allows for a robust, spatially-informed analysis of local resilience across Italy and offers a useful framework for identifying vulnerable or high-performing areas.

STEP 2: Social Growth Index (SGI)

The **Social Growth Index** is a composite indicator designed to measure the socio-economic resilience of a territory over time. It serves as a proxy for evaluating an area's ability to sustain development and absorb shocks. In our analysis, the index focuses on the growth of three key variables: Gross Domestic Product (GDP), Population, GDP per capita. In economic literature, these dimensions are interrelated through the following equation:

GDP = GDP per Capita × Population

From a mathematical perspective, annual growth is measured using the formula:

$$g(t) = \frac{V_{t+1} - V_t}{V_t},$$

where V_{t+1} *and* V_t represent the value of the variable at years $t+1$ and t, respectively. This provides a growth rate for each of the following indicators:

- $\frac{d(GDP)}{dt} = \%GDP$ – *Growth in total output*
- $\frac{d(GDP\ per\ Capita)}{dt} = \%GDP\ per\ Capita$– Growth in quality of output
- $\frac{d(Population)}{dt} = \%\ Population$ – Growth in quantity

This framework allows us to observe how each administrative unit evolves in terms of both size and well-being. The core assumption is that a higher capacity to maintain simultaneous growth in total output, per capita income, and population reflects a greater degree of socioeconomic resilience.

These variables were selected because they capture both the economic performance and demographic dynamics of a territory. Rather than combining them into a single numeric formula, we analyse their joint temporal growth trends and use comparative methods (such as the Copeland ranking) to assess and rank the resilience of different regions, provinces or municipalities.

2.1 Data Collection

We collected data from multiple sources to guarantee robustness with Eurostat serving as our primary provider.

Our initial dataset spanned 2000–2022 , reflecting the latest available figures. To address missing values in the European-region data, we supplemented Eurostat with World Bank Data records, achieving near-complete coverage across the full period. For Italian provinces, however, gaps persisted despite similar supplementation; consequently, we limited our analysis to 2015–2022.

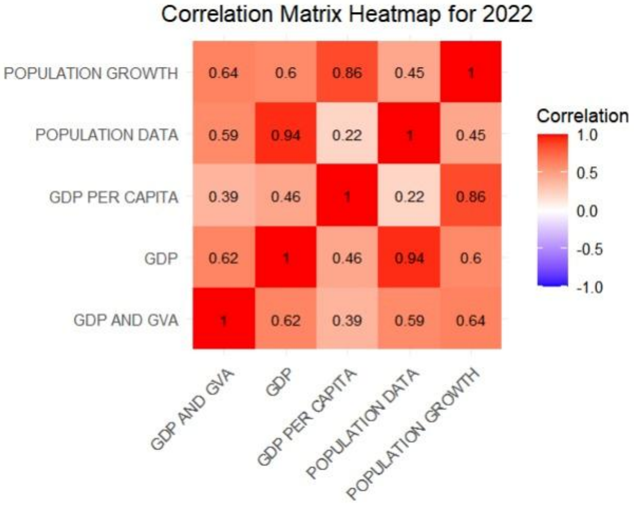
Because our principal aim is to assess the resilience of Italian municipalities, we required municipal-level GDP and population data. Population counts for each municipality (2015–2022) were obtained from ISTAT. In the absence of official GDP figures at this granularity, we estimated municipal GDP using the Fixed-Rank Kriging (FRK) method.

To support the application of this method and validate our municipal GDP estimates, we retrieved additional data from ISTAT, which includes employment figures at the municipal level, and from the MEF (Italian Ministry of Economy and Finance), which provides detailed financial data for local administrations.

2.2 Variable Redundancy Analysis

After collecting data from the various sources, we tested for redundancy among sources. If they convey the same content, it means some of them might be redundant and could be removed. To assess this, we examined the correlation between our variables. Fortunately, we

found that the pairwise correlations were very low, so we can proceed with the analysis using these variables.



STEP 3: LISA MAPS

To explore spatial dependence in the data, we computed Local Indicators of Spatial Association (LISA) for GDP, GDP per capita, and Population. The rationale behind this approach is that territorial units are likely interdependent: socio-economic conditions in one area may influence, or be influenced by, those of neighboring areas. If present, this spatial interdependence has important implications for both measurement and policy, as it implies that resilience cannot be properly assessed in isolation from geographic context.

LISA maps are based on the Local Moran's I statistic, which quantifies the similarity between each unit's value and that of its neighbors. For each year from 2000 to 2022, we applied the Univariate Local Moran's I to each variable separately, producing three outputs: p-value maps, cluster maps, and Moran scatter plots. The Moran scatter plot displays each region's standardized value (x-axis) against the spatial lag, i.e., the average standardized value of its neighbors (y-axis). The four resulting quadrants correspond to typical spatial configurations:

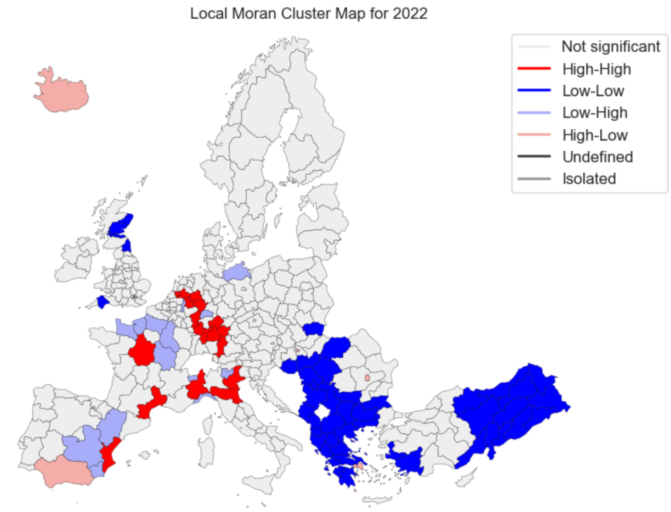
- High-High: units with above-average values surrounded by neighbors also above average
- Low-Low: units with below-average values surrounded by neighbors also below average
- Low-High: below-average units surrounded by above-average neighbors
- High-Low: above-average units surrounded by below-average neighbors

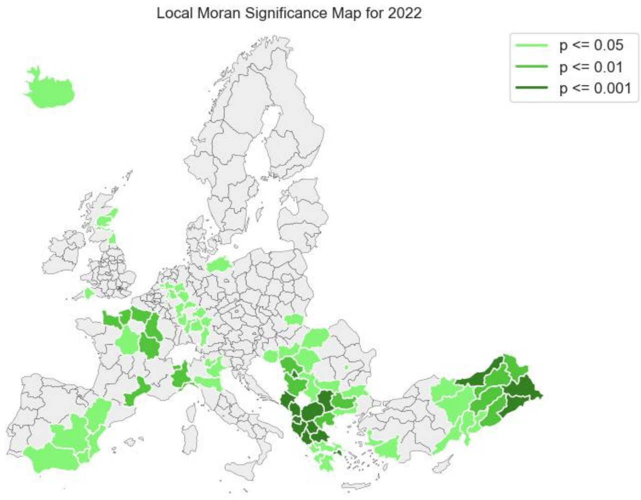
While the scatter plot visualizes all spatial associations, significance is assessed separately via permutation testing. The p-value and cluster maps identify which relationships are statistically significant (typically at $p < 0.05$), thus distinguishing meaningful patterns from random noise.

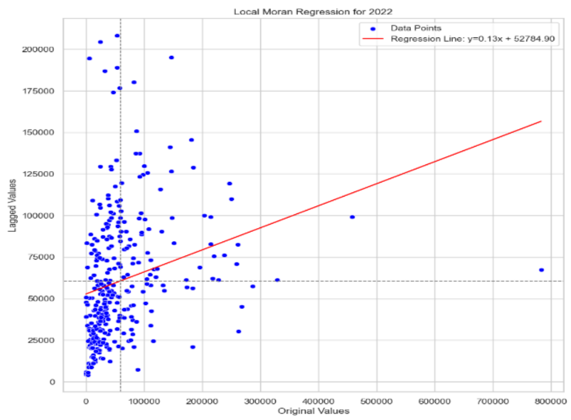
Results across years and variables consistently reveal strong positive spatial autocorrelation. Clusters of similar values—especially High-High and Low-Low—persist over time, suggesting underlying structural cohesion or shared socio-economic dynamics. Outlier configurations (High-Low and Low-High) offer additional insight into local deviations, which may be relevant for targeted policy analysis.

This evidence of spatial dependence is not only a key empirical result but also reinforces our methodological framework. It confirms that neighboring territories tend to influence each other, an assumption that is central to both the penalty-adjusted Copeland ranking and the application of Fixed-Rank Kriging (FRK) for estimating municipal-level GDP.

Incorporating spatial structure in both the evaluation (resilience ranking) and the estimation (GDP projection) steps ensures that our analysis accounts for inter-regional dynamics. In this way, the LISA analysis provides a strong statistical foundation for the use of spatially informed methods throughout the study of territorial resilience.







STEP 4: Copeland method on European regions and Italian provinces

After collecting data and selecting appropriate variables, we began evaluating the resilience of each province and region. To do so, we used the **Copeland method**, a ranking procedure based on pairwise comparisons: for each pair of territories, a "win" is assigned to the one with better performance, and a "loss" to the other. The final score for each unit is computed as the number of wins minus the number of losses.

$$S(i,k) = \sum_{l \in L} \begin{cases} +1, & a_{il} > a_{kl} \\ 0, & a_{il} = a_{kl} \\ -1, & a_{il} < a_{kl} \end{cases}$$

Using this method, we were able to assign a resilience score to each territorial unit (provinces and regions) and build a corresponding ranking. The evaluation was effective, as it highlighted the most resilient areas. However, we noticed a limitation: the resulting spatial pattern changed significantly over time. This instability was due to the sensitivity of the Copeland method to small data variations.

To address this issue, we introduced a **penalty term**. Specifically, we modified the value of each indicator (e.g., GDP) by combining the original value with the average value of neighbouring territories, using the formula:

$$P_i = w \cdot x_i - (1-w) \cdot \frac{1}{|N(i)|} \sum_{j \in N(i)} x_j$$

adjusted value = w × original value + (1 − w) × average of neighbours , where $0 \leq w \leq 1$

We can apply this method because we are under the assumption that neighbouring areas influence each other through cooperation and shared conditions. As shown in the LISA Maps being surrounded by strong territories can increase a unit's perceived resilience, and vice versa.

With this correction, we obtained a **more temporally consistent method**, reducing noise from small fluctuations. As a result, visible changes in the maps over time are more likely to reflect real geopolitical shifts rather than statistical artifacts.

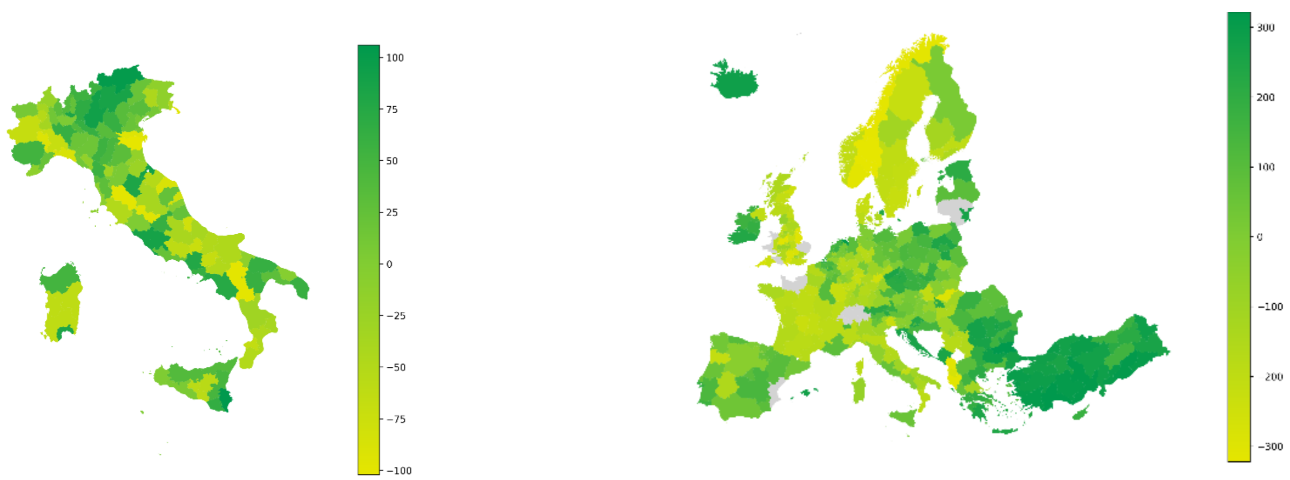


Figure: Copeland Ranking for Italian Provinces 2022 & Copeland Ranking for European regions 2022.

STEP 5: Estimating Municipal GDP via Fixed-Rank Kriging (FRK)

With the municipal GDP estimates generated via Fixed-Rank Kriging (FRK), we are now equipped to evaluate the resilience of each Italian municipality. Building on the earlier regional and provincial analysis, we apply the Copeland ranking method to compare municipalities across the three core dimensions: GDP growth, GDP per capita growth, and Population growth from 2015 to 2022.

Let $Y(s): s \in D$, where s represents GDP location within the spatial domain D . The general spatial model is specified as

$$Y(s) = t(s)^{\top\alpha} + \nu(s) + \xi(s), \quad s \in D,$$

Where:

- $t(s)$ is a vector of observed covariates (e.g., population, employment, financial indicators),
- α is a vector of regression coefficients,
- $\nu(s)$ is a spatially structured random effect capturing medium- to large-scale spatial correlation
- $\xi(s)$ is a fine-scale spatial error term, assumed to be spatially uncorrelated.

We assume that $E[\nu(\cdot)] = 0$ and $E[\xi(\cdot)] = 0$. Define the combined process $\lambda(\cdot) = \nu(\cdot) + \xi(\cdot)$, which implies $E[\lambda(\cdot)] = 0$. This formulation captures both the systematic effects of known predictors and residual spatial heterogeneity that cannot be explained through covariates alone.

5.1 Spatial Random Effects Representation

To model $\nu(\cdot)$ we use a **Spatial Random Effects (SRE)** representation:

$$\lambda(s) = \sum_{\ell=1}^{r} \phi_l(s) n_l, \quad s \in D,$$

Where:

- $\eta_1, \ldots, \eta_r$ is a random vector of dimension r
- $\phi_1(\cdot), \ldots, \phi_r(\cdot)$ are predefined spatial basis functions, often chosen to span multiple resolutions

5.2 Application and covariates

We applied the FRK methodology using province-level GDP data as the observed values and municipal-level covariates as predictors. These include:

- **Population size and growth** (ISTAT),
- **Number of employed individuals by type of economic activity** (ISTAT),
- **Municipal fiscal capacity and expenditures** (MEF).

These covariates were selected based on their economic interpretability and spatial completeness. The estimation process leverages both the spatial configuration of the municipalities and the functional relationship between covariates and GDP.

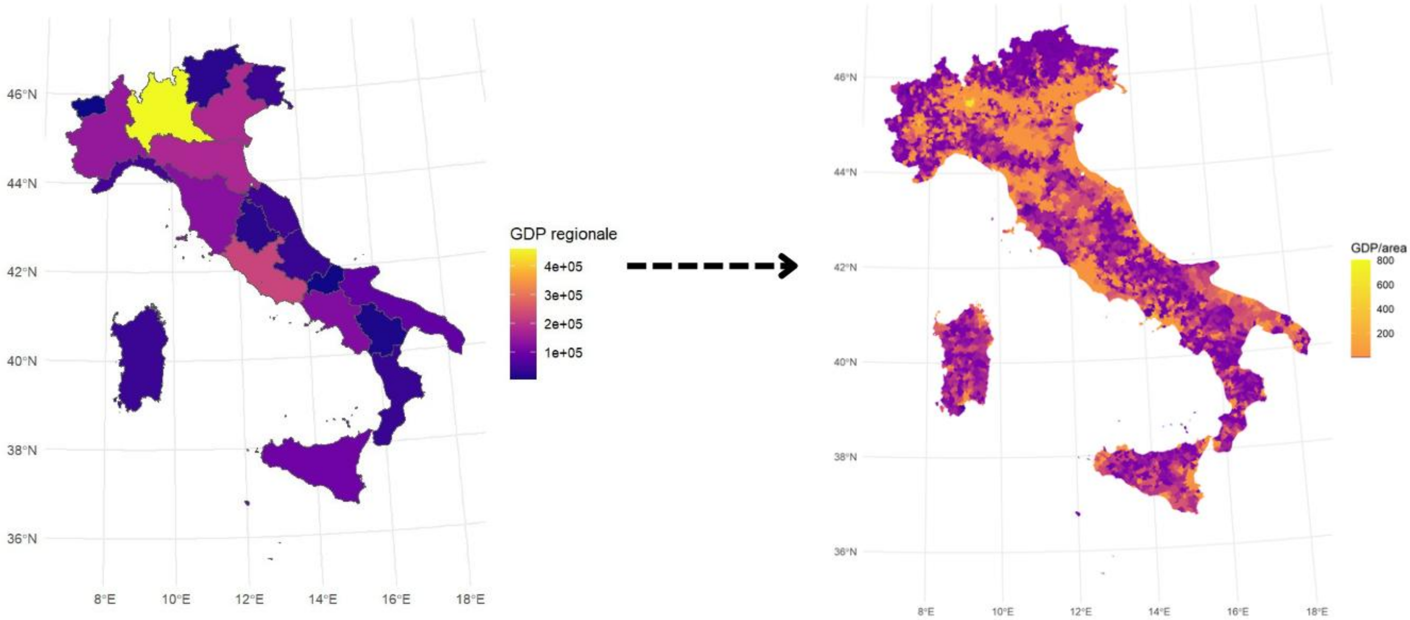


Figure: Fixed Ranking Kriging Applied to GDP regionale to map municipalities GDP.

These spatial patterns validate the FRK model's ability to capture broad and fine-scale spatial dynamics. Furthermore, this mapping provides the essential input for municipal-level resilience assessment, enabling a more granular understanding of territorial disparities than regional aggregates alone.

The decision to use FRK is justified not only by computational tractability but also by its alignment with our overall methodology. The spatial smoothing inherent in FRK respects the same assumptions of spatial dependence demonstrated by the LISA analysis. Moreover, it allows us to reconstruct unobserved GDP values in a statistically principled way, informed by neighbouring areas and relevant socio-economic indicators.

This step is crucial in bridging the data gap at the municipal level and ensures that our subsequent application of the Copeland method is grounded in reliable and spatially coherent economic estimates.

STEP 6: Resilience Ranking at the Municipal Level Using the Copeland Method

With the municipal GDP estimates generated via Fixed-Rank Kriging (FRK), we are now equipped to evaluate the resilience of each Italian municipality. Building on the earlier regional and provincial analysis, we apply the Copeland ranking method to compare municipalities across the three core dimensions: GDP growth, GDP per capita growth, and Population growth from 2015 to 2022.

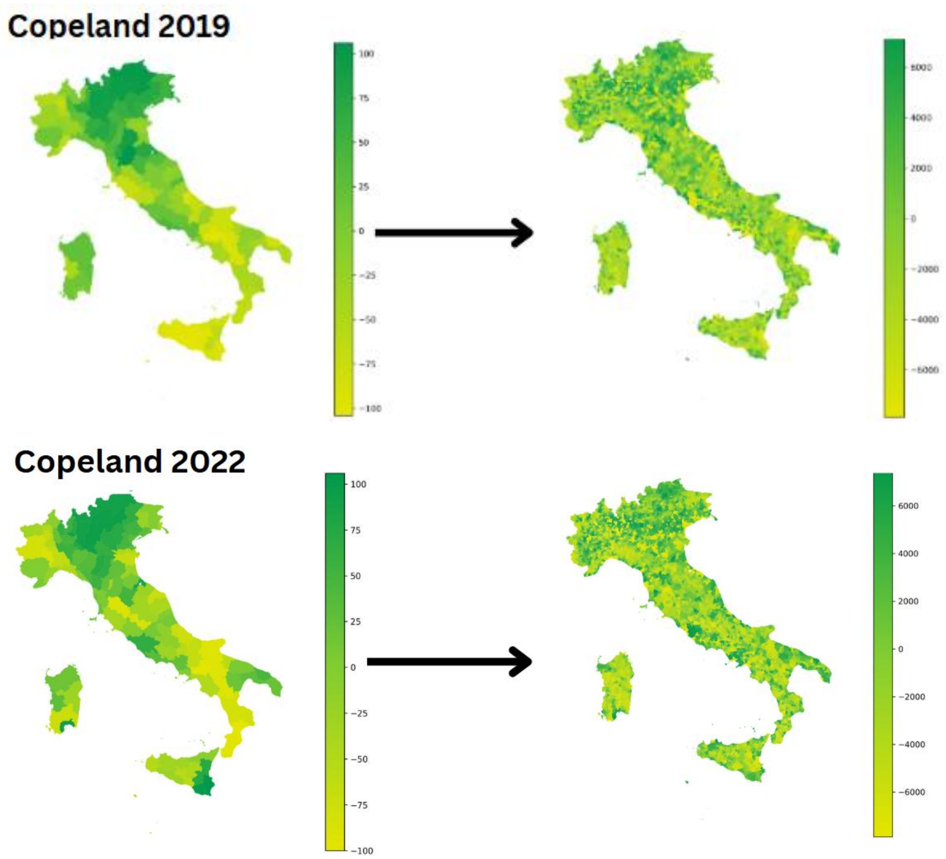


Figure: Copeland Methodology Applied to regions and the corresponding maps to regional and municipals levels

6.2 Copeland Score changes

In the figure below we have seen that in many municipalities the Copeland score changes. Many regions in the south tend to see a decrease in Copeland score while those in the north see an increase.

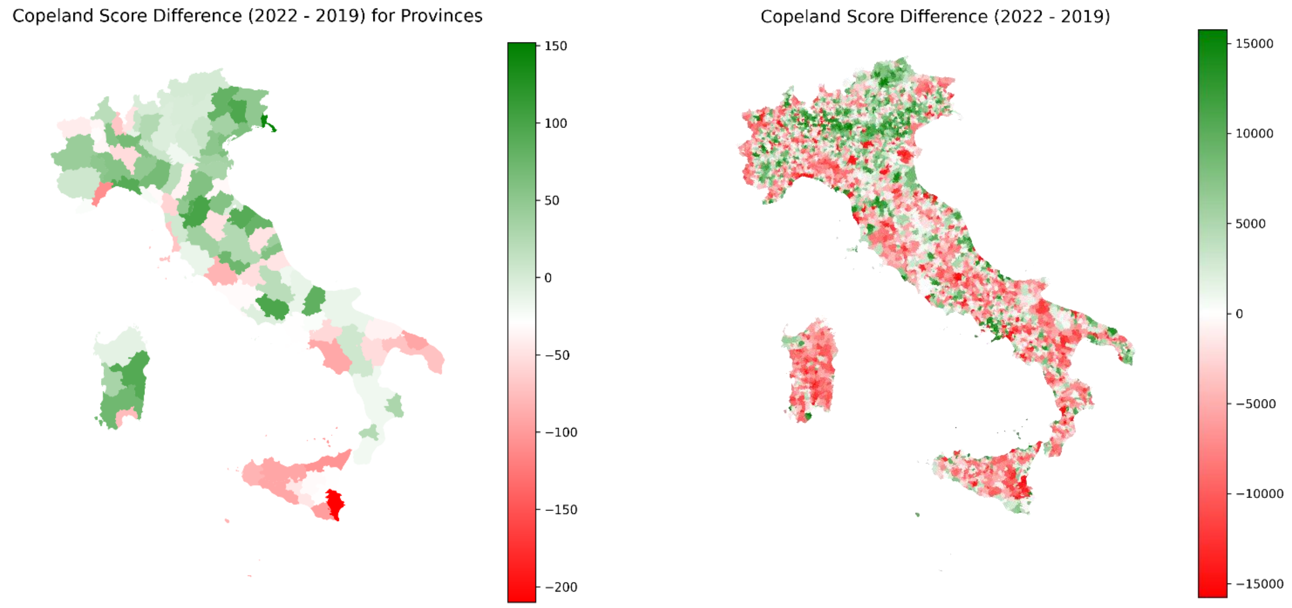


Figure: Copeland Score changes by municipalities level and provinces

Compared to regional or provincial rankings, the municipal-level Copeland assessment:

- Captures fine-scale variation, revealing local strengths and vulnerabilities,
- Supports targeted policymaking, especially in regions with uneven development,
- Aligns with administrative realities, as many social and economic policies are implemented at the municipal level.

By combining a robust spatial estimation of GDP with a comparative resilience framework (this step), we provide a comprehensive and spatially informed measure of territorial resilience across Italy.